Pietro Zunino

Ph.D. Candidate

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EXPERIENCE

Ph.D. Candidate — **Jan 2025 - Present**
Technical University of Denmark (DTU) — *Roskilde, Denmark*

- Contributing to the DTEC GREAT project to develop realistic EV charging strategies and analyze the provision of TSO services from EVs, evaluating their effects on the LV grid.

Research Assistant — **May 2024 - Dec 2024**
Technical University of Denmark (DTU) — *Roskilde, Denmark*

- Contributed to the EV4EU project by testing EV smart charging strategies and ancillary services using experimental campus chargers, including hardware programming via BeagleBones.

EDUCATION

Politecnico di Milano, Italy | *Master's Degree* — **2021 - 2024**

- Grade: 110L/110 (summa cum laude).
- Master Thesis as an Exchange Student at DTU (Sep 2023 - Feb 2024): Analysis and testing of frequency services provided by electric vehicle clusters.

Università degli Studi di Genova, Italy | *Bachelor's Degree* — **2018 - 2021**

- Grade: 110L/110 (summa cum laude).
- Bachelor Thesis: Frequency controller for multi-area power systems using Hardware in the Loop simulations.

PUBLICATIONS

- **Zunino, P.**, Marinelli, M., Erel, B. & Engelhardt, J. (2026). Mitigating Synchronization Effects in Price-Aware EV Charging Using Equitable, Communication-Free V-P Control. *10th E-Mobility Power System Integration Symposium*, Porto, Portugal. [Accepted, to be presented].
- **Zunino, P.**, Unterluggauer, T., Marinelli, M. & Engelhardt, J. (2026). Impact of electric vehicles on low-voltage distribution networks: A critical review. *eTransportation*, 28, 100594.
- **Zunino, P.**, Engelhardt, J., Striani, S., Pedersen, K. L. & Marinelli, M. (2025). Frequency Control in EV Clusters: Experimental Validation and Time Response Analysis of Centralized and Distributed Architectures. *Proceedings of 2024 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE).*

ACTIVITIES & AWARDS

- **IDEA League Challenge Programme:** Selective program on the energy transition impact, across five European universities (ETH Zurich, Polimi, TU Delft, RWTH Aachen, and Chalmers).
- **ENHANCE Summer School (Chalmers University):** Mobility program focused on sustainable entrepreneurship and collaborative problem-solving in international teams.
- **University Startup Challenge Winners:** Co-developed *RECreate*, an app to facilitate the creation of Renewable Energy Communities (Politecnico di Milano & Bocconi).
- **2020 IRIS & Hitachi Rail Awards:** Awarded to top-performing Bachelor's students at Università degli Studi di Genova.

SKILLS

Technical Grid modelling (Pandapower), Monte Carlo simulations, Matlab & Simulink, BeagleBone

Expertise Electric Vehicles & Smart Charging, Vehicle-to-Grid (V2G), Ancillary Services, Distributed Energy Resources, Renewable Energy, Power Electronics, Electrical Machines, EMC